ViewTouch Label Utility

User Manual

Shelf, price, cigar and receiving labels for ViewTouch POS

Raspberry Pi label station · Zebra label printers

Version 1.1 · November 2026



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1. How the program works

ViewTouch Label Utility prints shelf, price, cigar-stick and receiving labels for the items in your ViewTouch POS. It runs on the Raspberry Pi next to ViewTouch and sends labels to your Zebra label printers.

What it does, in order

1. **Reads your items from ViewTouch.** Names, prices, UPCs (including alternate and carton UPCs), brands and categories come straight from ViewTouch's database. The program only *reads* it — nothing you do here changes an item in ViewTouch.
2. **Uses a label template.** A template is a saved label design: its size, the printer it goes to, and where each piece (description, price, barcode, date…) sits on the label.
3. **Draws the whole label as one picture** at the printer's own resolution (203 dots per inch on most Zebras). Text and barcodes are drawn by the program, bar by bar, so every bar is an exact number of printer dots wide.
4. **Sends that picture to the printer.** The printer only prints the dots it is given; it never substitutes its own fonts or barcodes. This is the same way the old ILSA system printed, which is why labels look identical every time.

> **Good to know:** What you see on the print-preview screen is drawn from the same layout as the printout. Double-click the preview to see the exact printer dots.

Where things are kept

What	Where
The program	`~/viewtouch-main/modern/tools/vt_label_utility.py`
Your label templates (live copy)	`~/.local/share/ViewTouch/label_templates.json`
Starting templates shipped with the program	`~/viewtouch-main/modern/seed/templates/label_templates.json`
Items, prices, UPCs, printers	ViewTouch's own database (read only)
Error log	`/tmp/vt_label_utility.log` (cleared when the Pi restarts)
Last label sent to a printer	`/tmp/last_label.zpl` and `/tmp/VT_Labels.pdf`

The label utility keeps its own template list, so templates you delete stay deleted.

2. Starting and closing

- **From ViewTouch:** on the Manager screen, press the **Label Utility** button. While the label utility is open, ViewTouch will not log you out for being idle.
- **For one item:** if ViewTouch opens the label utility for a particular item (a button set up with the `openLabelUtility` command and a UPC), it goes straight to that item's label preview. Any of the item's codes works — primary, alternate or carton UPC.
- The program opens full-screen. To close it, press **Close** (red, bottom-left).
- The title bar says *ViewTouch Label Utility v1.1*; the start screen shows how many ViewTouch items it loaded.

> **Note:** Item changes made in ViewTouch (new items, price changes) appear here the next time the program opens, or straight away if you press **Reload viewtouch.db**.

The four screens

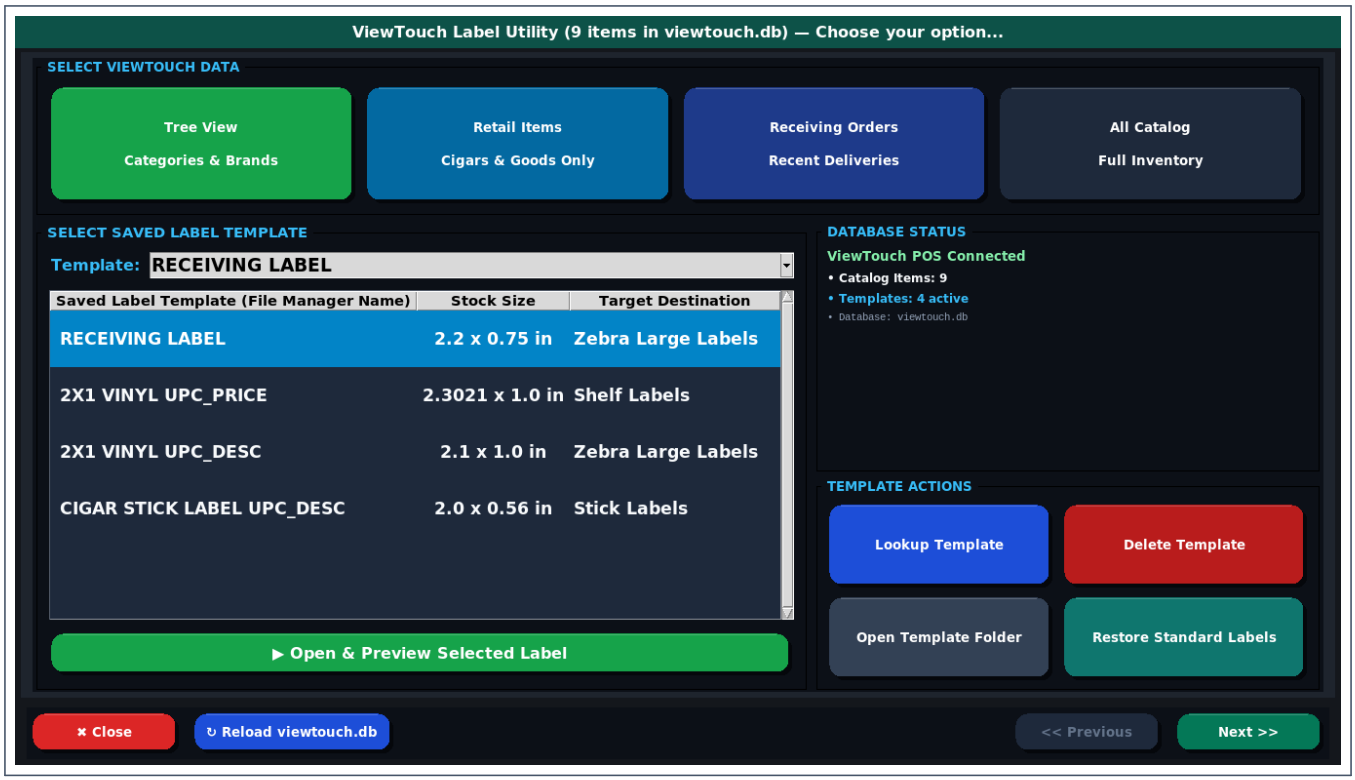
The program works like a wizard. You move forward with **Next >>** and back with **<< Previous** (bottom of the window).

Screen	What you do there
1. Start	Choose which items to work with, or jump straight to a saved label template.
2. Categories	Narrow the items down by group, category or brand (Tree View only).
3. Items	Tick the items to print, set quantities, and choose which UPC goes on each label.
4. Preview & Print	Pick the template, check the preview, edit the label design, and print.

Bottom bar (always visible)

Button	What it does
Close	Closes the program.
Reload viewtouch.db	Re-reads all items from ViewTouch (use after changing items in ViewTouch). This also resets any UPC choices made on screen 3.
<< Previous	Goes back one screen.
Next >>	Goes forward one screen.

3. Screen 1 — Start



Screen 1 — choose items or a saved template

Select ViewTouch data (top row)

Each big button chooses which items you will work with and moves you on immediately.

Button	Shows	Goes to
Tree View — Categories & Brands	All items, organised by group, category and brand so you can pick a section.	Screen 2
Retail Items — Cigars & Goods Only	Only retail items (items with a real barcode, merchandise, or cigar/tobacco/retail categories).	Screen 3
Receiving Orders	Items sold in packs or cartons, plus retail items — handy after a delivery.	Screen 3
All Catalog — Full Inventory	Every item.	Screen 3

Select saved label template (left)

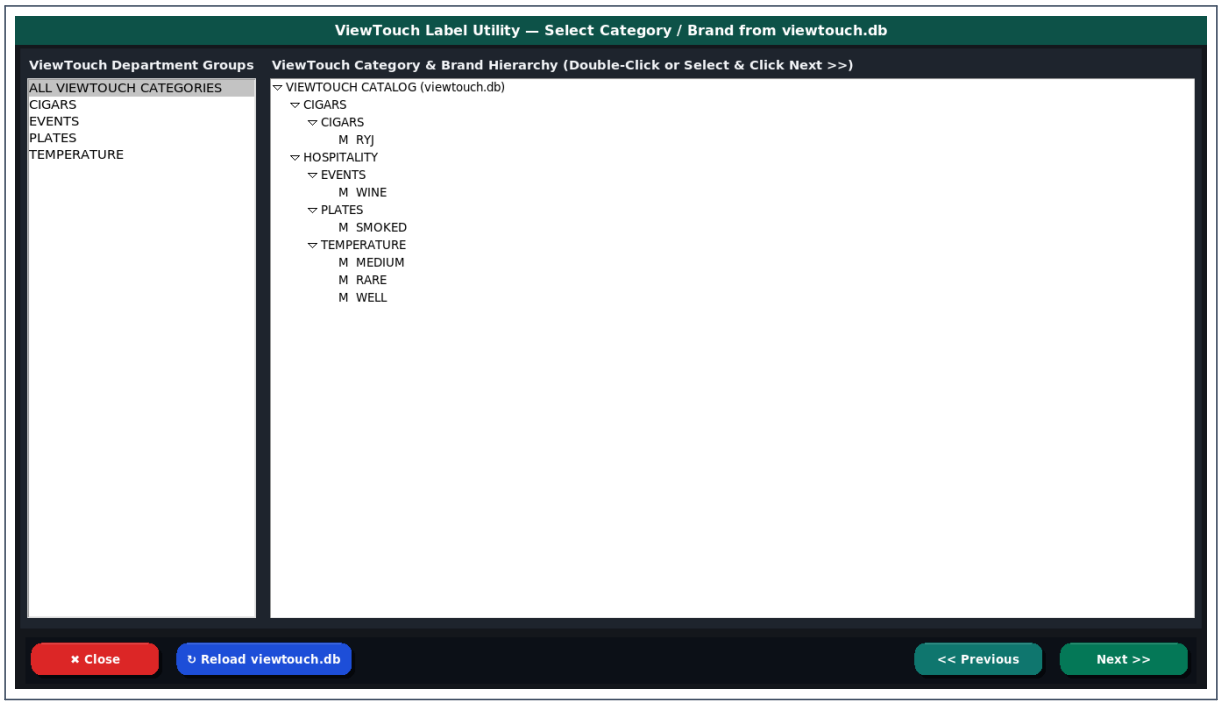
Lists every label template. To go straight to a label's preview:

- Tap a template in the list (or choose it in the **Template** drop-down), then press **Open & Preview Selected Label** — or simply double-click the template.
- This opens screen 4 with that template. If no items have been ticked yet, the preview uses the first item in the list. This is the quickest way to print a **RECEIVING LABEL**, which does not need an item.

Database status and template actions (right)

Item	What it means / does
Database status	Confirms ViewTouch is connected and shows how many items and templates were loaded.
Lookup Template	Imports a label template from a file — including ILSA **.rtm** labels. See section 9.
Open Template Folder	Opens the `seed/templates` folder in the file manager — a handy place to keep files you want to import.
Delete Template	Deletes the template selected in the list on the left (it asks first). The last remaining template can't be deleted.
Restore Standard Labels	Puts the four standard labels (RECEIVING LABEL and the three ILSA labels) back to their original layout and printer, replacing any changes made to them. Your other templates are kept. It asks first.

4. Screen 2 — Categories (Tree View)

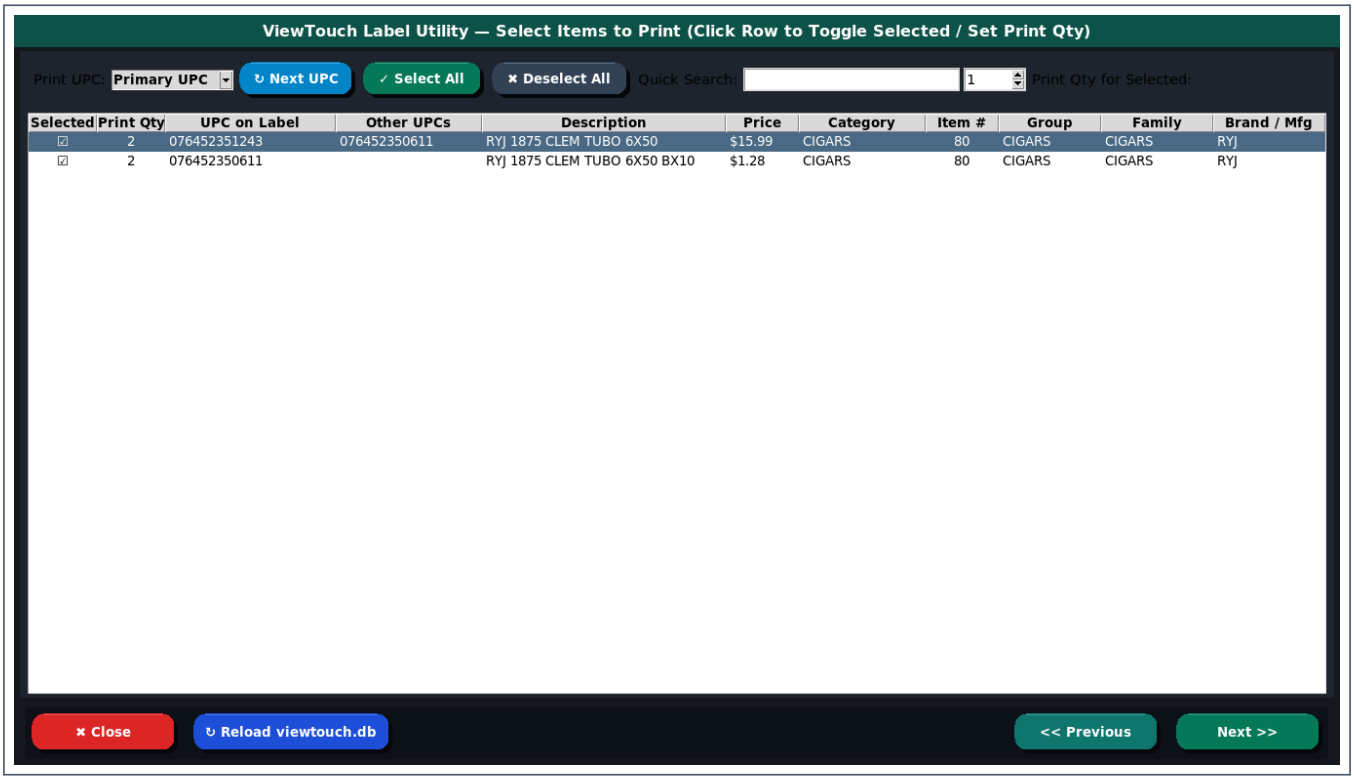


Screen 2 — groups, categories and brands

- **Left list — ViewTouch Department Groups:** tap a category to show only that category in the tree. **ALL VIEWTOUCH CATEGORIES** shows everything.
- **Right — Category & Brand Hierarchy:** groups (Cigars, Retail…) contain categories, which contain brands (shown with an **M** in front).
- Tap a group, category or brand and press **Next >>** — or just double-click it — to list only those items on screen 3. Selecting the top line (or nothing) lists every item.

5. Screen 3 — Items

This screen lists the items. Tick the ones you want labels for.



Screen 3 — two items ticked; the highlighted row will print its primary UPC

Selecting items

- **Tap a row** to tick it (the empty box in Selected gets a tick). Tap again to untick. Ticking a row also sets its **Print Qty** to the number in **Print Qty for Selected**.
- **Select All** / **Deselect All** tick or untick every item in the list.
- **Quick Search** filters the list as you type. It matches description, brand, category and *any* of an item's UPCs — so you can scan or type an alternate UPC to find an item.
- **Print Qty for Selected** (right) sets the quantity on every ticked item.

> **Quantities:** The number you enter in the print window (section 7.2) is printed for **every** ticked item. The Print Qty column only sets the number the print window starts with — items can't have different quantities in the same print run. To print different amounts, print the items in separate runs.

- Press **Next >>** to go to the preview with the ticked items. If nothing is ticked, the first item is used.

Columns

Column	Meaning
Selected / Print Qty	Whether the item is ticked and its quantity.
UPC on Label	The code that will be printed. If it is not the primary UPC, it is tagged, e.g. *04252614 [Alt UPC 1]*.
Other UPCs	The item's other codes you could print instead.
Description, Price, Category, Item #, Group, Family, Brand / Mfg	Item details from ViewTouch.

Choosing which UPC goes on the label

Many items have more than one code in ViewTouch — a primary UPC, alternate UPCs, and for packs the carton UPC. You can print any of them. **This only affects the label; it never changes the item in ViewTouch.**

Control	What it does
Print UPC (drop-down)	Switches the ticked items (or every listed item if none are ticked) to **Primary UPC**, **Alt UPC 1–3** or **Carton UPC**. Items that don't have that code keep their primary UPC.
Next UPC	Steps the highlighted row (or the ticked rows) to its next code: Primary, Alt 1, Alt 2…, Carton, then back to Primary.

> **Note:** The chosen code is used for both the barcode and the printed UPC number on the label. Choices reset to the primary UPC when you press Reload viewtouch.db or restart the program.

6. Screen 4 — Preview & Print

The large area shows the label exactly as it will print, for one item at a time.



Screen 4 — toolbar (wraps onto a second row on the Pi's screen) and the label preview

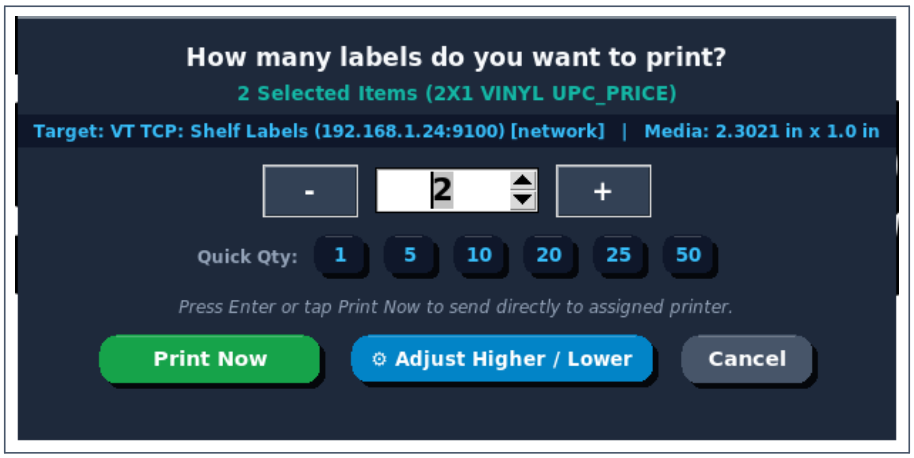
Control	What it does
Template	Chooses which label design to use. The preview updates immediately.
Print Labels	Opens the print window (section 7).
Arrow buttons and 1/5	Steps through the ticked items. The number shows which item is previewed.
UPC	Lists every code for the previewed item (e.g. *Alt UPC 1: 04252614*). Pick one to print that code on this item's label.
Edit Report	Opens the label designer for the current template (section 8).
Create Report	Makes a new template as a copy of the current one (section 8.6).
Printer Offset	Printer position, darkness, resolution and calibration for this template (section 7.3).
Printer badge	Shows which printer and label size this template uses.
Double-click the preview	Switches between the sharp preview and the exact printer-dot view. The caption under the label says which one you are seeing.

7. Printing labels

7.1 Step by step

1. On screen 1 choose how to find your items (e.g. **Retail Items**), or use **Tree View** and pick a brand.
2. On screen 3 tick the items. Use **Print UPC** or **Next UPC** if a label needs an alternate or carton UPC.
3. Press **Next >>**. On screen 4 choose the **Template** (e.g. *2X1 VINYL UPC_PRICE*).
4. Check the preview. Use the arrow buttons either side of the item counter to look at each item.
5. Press **Print Labels**. If the template has a Variable field, you are first asked for the text (press OK, or Skip).
6. Enter **how many labels** per item and press **Print Now** (or Enter). A green message confirms the labels were sent.

7.2 The print window



The print window

Control	What it does
How many labels?	Number of labels printed **for each selected item**. Use - and +, type a number, or tap a Quick Qty button (1, 5, 10, 20, 25, 50). It starts at the first item's Print Qty.
Target / Media line	The printer and label size the job will go to. Check this before printing.
Print Now	Sends the labels. Pressing Enter does the same.
Adjust Higher / Lower	Closes this window and opens Printer Offset (section 7.3).
Cancel	Closes without printing.

> **Example:** 3 items ticked and 2 entered here prints 6 labels — 2 of each item.

Where labels go

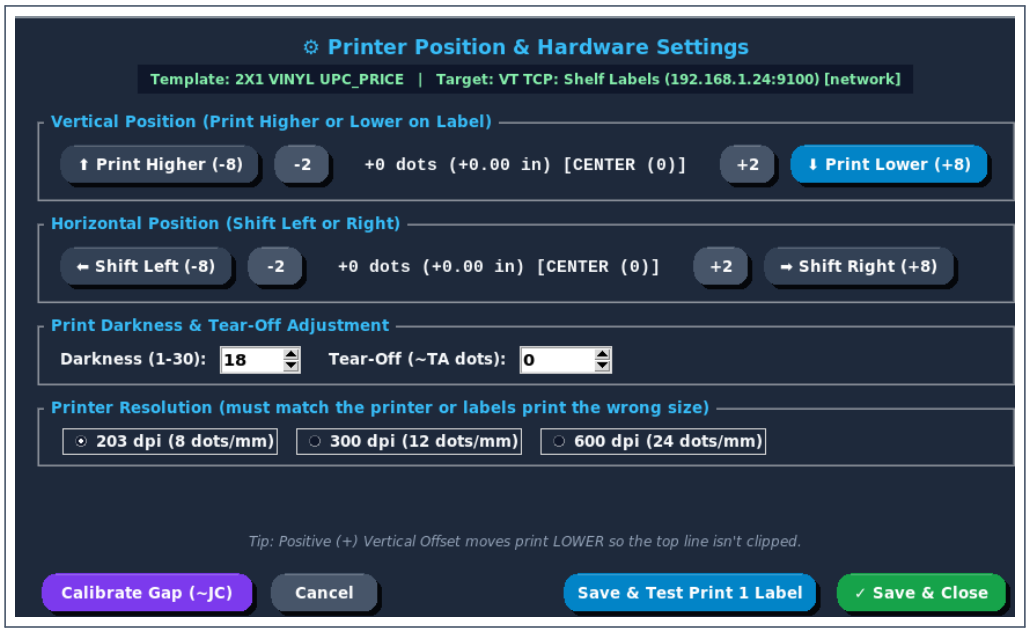
Each template remembers its printer. The printers come from ViewTouch's printer settings.

Printer type	How the label is sent
Network label printer (e.g. Zebra Large Labels, Shelf Labels, Stick Labels)	Sent directly to the printer's IP address as a ZPL picture.
CUPS printer queue	Label printers get the same ZPL picture; ordinary printers get a PDF page the size of the label.
PDF Preview (Desktop)	Opens a PDF of the labels on screen instead of printing.
Receipt printers (Epson, receipt, kitchen…)	Get a plain text version. Don't use these for labels.

> **Changing a printer's IP:** Change it in ViewTouch's printer settings. Templates follow the printer by name, so a template set to **Stick Labels** automatically uses Stick Labels' new address.

7.3 Printer Offset (printer settings)

Open with **Printer Offset** on screen 4 or **Adjust Higher / Lower** in the print window. Settings are saved **per template**.



Printer Offset — printer position, darkness, resolution and calibration

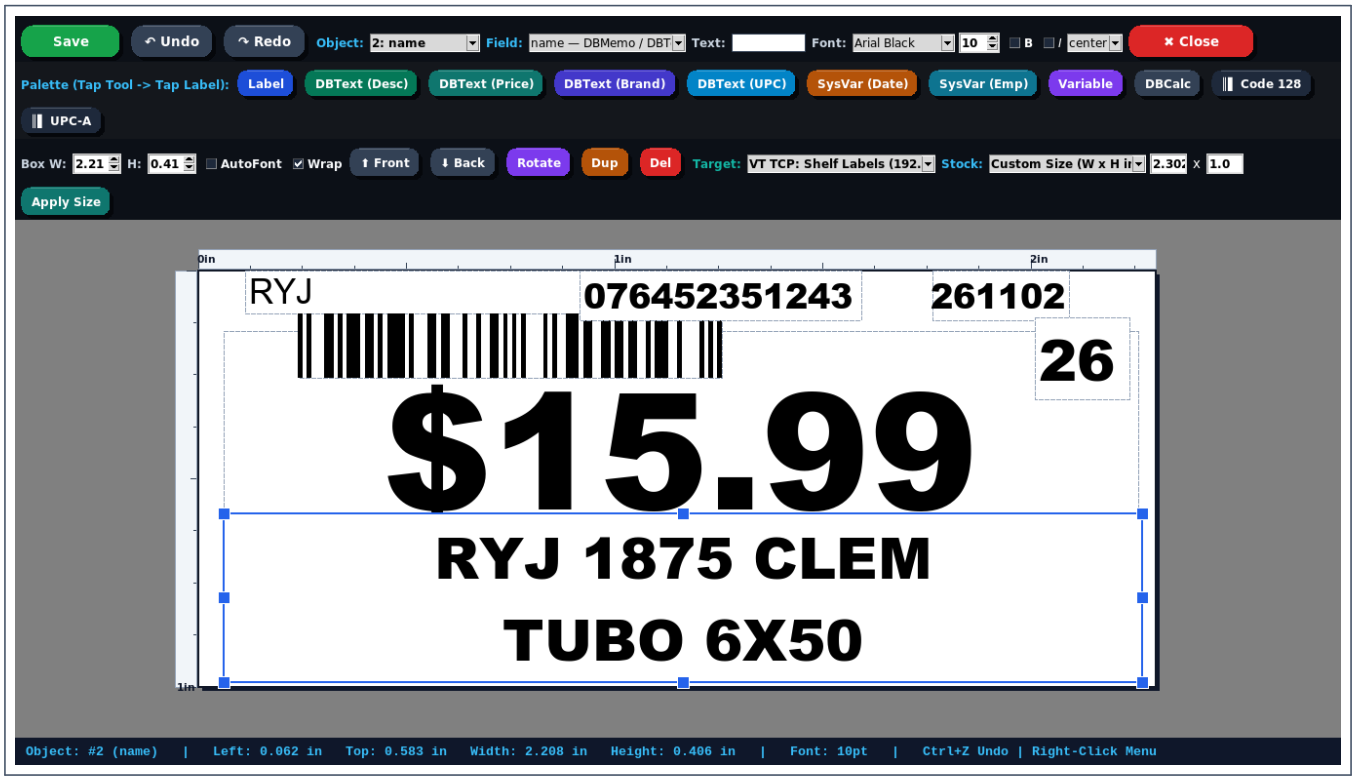
Setting	What it does
Vertical Position	**Print Higher (-8)** / **Print Lower (+8)** move everything up or down by 8 dots (about 1 mm). -2 / +2 make finer moves. Use this if the top or bottom line is cut off.
Horizontal Position	**Shift Left** / **Shift Right** move everything sideways, the same way.
Darkness (1–30)	Higher is darker. Raise it if bars or text look faint; lower it if bars bleed together.
Tear-Off	Adjusts where the label stops for tearing off. Leave at 0 unless labels tear in the wrong place.
Printer Resolution	203, 300 or 600 dpi. **Must match the printer** (shown on the printer's label) or labels print the wrong size.

Setting	What it does
Save & Close	Saves and closes.
Save & Test Print 1 Label	Saves and prints one label of the previewed item, so you can check the position.
Calibrate Gap	Makes the printer feed a few labels to measure the label length. Do this after loading a new roll or a different label size.
Cancel	Closes without saving.

Note: Test Print and Calibrate Gap only work with network printers (ones with an IP address).

8. Editing labels (the label designer)

Press **Edit Report** on screen 4. The designer opens the current template, filled in with the item being previewed, so you can see real data while you work.



The label designer — three toolbars above the label; the description is selected (blue box and handles)

8.1 Moving and sizing pieces with the mouse or touch

- **Click** a piece (description, price, barcode…) to select it. A blue dashed box with 8 handles appears.
- **Drag** the piece to move it. **Drag a handle** to change its box size.
- **Ctrl-click** or **Shift-click** to select several pieces; **drag on an empty area** to draw a selection box around them.
- **Double-click** a piece for its Properties window (field, text, exact position, rotation).
- **Right-click** for a menu: Add Field, Bring to Front, Send to Back, Position / Properties, Duplicate, Rotate 90°, Delete Component.

Properties window (double-click a piece)

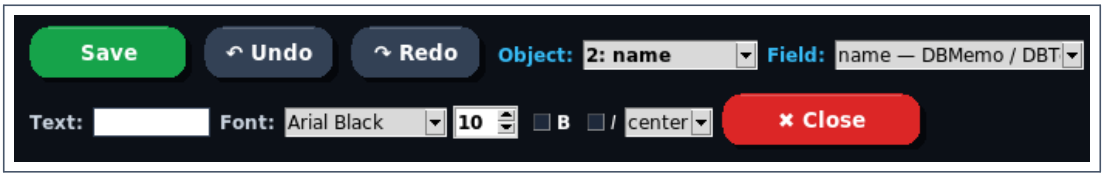
Field	What it does
Data Field / Type	What the piece shows — the same list as the Field box in the top toolbar (section 8.5).
Custom / Static Text	The words for a Label or Variable piece.
Left / Top (in)	The exact position of the piece's top-left corner, in inches from the label's top-left corner.
Rotation (deg)	0, 90, 180 or 270.
Save Properties / Cancel	Apply the changes, or close without changing anything.

Status bar (bottom of the designer)

Shows the selected piece's exact size and position, updated as you drag: `Object: #2 (name) | Left: 0.062 in Top: 0.583 in Width: 2.208 in Height: 0.406 in | Font: 10pt`.

> **Fine-tuning small labels:** For millimetre-critical labels such as the cigar stick label, use the status bar together with the arrow keys (0.01 in per press, 0.05 in with Shift), **Left / Top** in the Properties window, and **Box W / H** in the second toolbar for exact width and height. 0.01 in is about 0.25 mm, or 2 printer dots at 203 dpi.

8.2 Top toolbar



Top toolbar (shown in two parts)

Control	What it does
Save	Saves the template and closes the designer.
Undo / Redo	Steps back or forward through your changes.
Object	Picks a piece by name (useful when pieces overlap).
Field	Changes what the selected piece shows (see the field list in 8.5).
Text	The words for a static Label or Variable piece. Press Enter to apply.
Font / size / B / I	Font (Sans, Serif, Mono, Arial, Arial Black, Courier New), point size, bold, italic.
Alignment	left, center or right inside the piece's box.
Close	Closes the designer. If there are unsaved changes it asks: **Yes, Save**, **No, Discard** or **Cancel**.

8.3 Palette — adding new pieces

Tap a palette button, then tap on the label where the new piece should go. Press **Esc** to cancel.



Palette (shown row by row)

Palette button	Adds
Label	Fixed text you type (e.g. SALE).
DBText (Desc) / (Price) / (Brand) / (UPC)	The item's description, price, brand or UPC number.
SysVar (Date) / SysVar (Emp)	Today's date (YYMMDD) or the employee's name as F LAST.
Variable	Text you are asked for each time you print (e.g. RECEIVED).
DBCalc	Pack calculation, e.g. PK:10 ($1.20/EA).
Code 128 / UPC-A	A barcode of the item's UPC (the one chosen on screen 3 or 4).

8.4 Second toolbar



Second toolbar (shown row by row)

Control	What it does
Box W / H	Exact width and height of the selected piece, in inches.
AutoFont	On: text shrinks or grows to fill its box. Off: uses the exact font size you set (the ILSA labels use exact sizes).
Wrap	Lets long text use two lines.
Front / Back	Brings the piece in front of, or sends it behind, overlapping pieces.
Rotate	Turns the piece 90°.
Dup / Del	Duplicates or deletes the selected piece.
Target	The printer this template prints to.
Stock	Picks a standard label size (2.0×1.0, 2.5×0.75, 3.0×2.0, 1.5×1.0, 2.25×1.25) or Custom.
W × H + Apply Size	Type an exact label size in inches and press Apply Size.

Keyboard shortcuts

Keys	Action	Keys	Action
Ctrl+Z / Ctrl+Y	Undo / Redo	Arrow keys	Nudge 0.01 in
Ctrl+C / X / V	Copy / Cut / Paste	Shift+Arrows	Nudge 0.05 in
Ctrl+D	Duplicate	R	Rotate 90°
Ctrl+A	Select all pieces	Delete	Delete selected
Esc	Cancel palette tool		

8.5 Field list

Field	Prints
Label (Static Text)	The text you type.
Variable	Text asked for when printing (default RECEIVED).
Date (YYMMDD) / Date (MM/DD/YYYY) / Year (YY)	Today's date in that format.
Employee (F LAST)	The active ViewTouch employee's name, e.g. P SMITH.
Description, Retail Price, Manufacturer / Brand, Category, UPC / PLU	The item's details from ViewTouch.
DBCalc	Pack quantity and per-unit price.
Code 128 / UPC-A / UPC-E / Code 39	A barcode. UPC fields automatically use the right type for the code chosen (UPC-A, UPC-E, EAN-13), and the check digit is always calculated correctly.

ILSA labels: Labels imported from ILSA keep ILSA's bar width, the digits printed under the bars, and their exact text placement. These settings are carried over automatically and are not shown in the designer.

8.6 Creating a new label (Create Report)

1. On screen 4 choose the template closest to what you want, then press **Create Report**.
2. Enter a **Label / Store Name** (this becomes the template name).
3. Choose the **Target Printer** and the **Media Stock / Size**.
4. Press **Create & Open Designer**. The new template is a copy of the one you started from, ready to edit.

9. Importing labels

Use **Lookup Template** on screen 1. A file window opens; choose the file and press Open.

ILSA labels (.rtm)

Label designs saved from the old ILSA system (ReportBuilder **.rtm** files) import directly. The program reads the label size, the position and size of every piece, fonts, alignment, barcode type, bar width and the digits under the barcode, and creates a template with the same name as the file.

1. Copy the .rtm file to the Pi (a USB stick, or put it in the folder that **Open Template Folder** opens).
2. Press **Lookup Template** and choose the file.
3. If a template with that name already exists you are asked whether to replace it.
4. A message confirms the import. The new label opens in the preview.
5. Open **Printer Offset**: set the right printer, check the resolution, and print a test label.

Fields understood in ILSA files: description (PLU_Desc), UPC (PLU_Num), price, manufacturer, category, dates (yymmdd, yy, mm/dd/yyyy), fixed text labels, and UPC-A, UPC-E, EAN-13, Code 39 and Code 128 barcodes.

After importing: ILSA saved some labels a little wider than the actual stock (for example 2.3 in for a 2 in label). This is kept as-is so the layout matches ILSA. If the right edge is cut off, adjust the size with W × H + Apply Size in the designer.

Other files

File	Result
.json	Template file(s) from this program or the other ViewTouch label programs. Every template in the file is added.
.prn, .nlbl, .slnx	ZebraDesigner / NiceLabel / BarTender files can't be read. A new template with the file's name is created as a copy of RECEIVING LABEL for you to design in the designer.

10. Your label templates

These four templates are installed. They match the labels from the ILSA system.

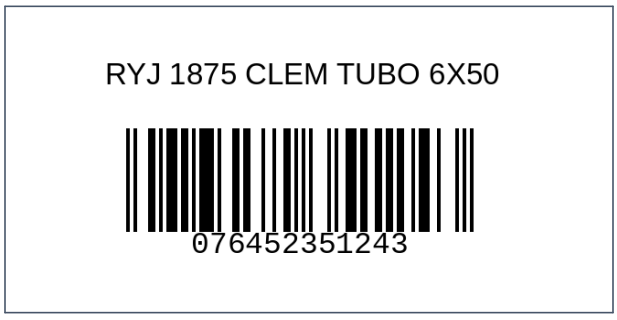
2X1 VINYL UPC_PRICE — shelf price label

Brand, UPC, date (YYMMDD), barcode, year code, large price in Arial Black, and description. Printer: Shelf Labels.



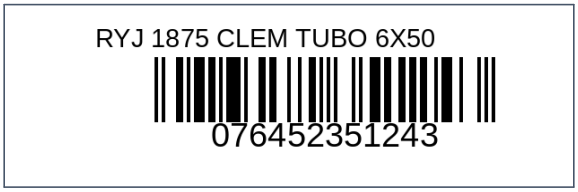
2X1 VINYL UPC_DESC — description and barcode

Description with a full-size UPC-A barcode and the digits underneath. Printer: Zebra Large Labels.



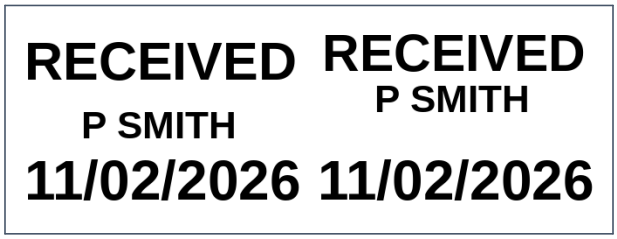
CIGAR STICK LABEL UPC_DESC — 2.0 × 0.56 in

Description and UPC-A barcode for cigar sticks. Printer: Stick Labels (the cigar printer at 192.168.1.25).



RECEIVING LABEL — 2.2 × 0.75 in, two across

RECEIVED, the employee's name and today's date, twice side by side. Needs no item.



Deleting and restoring templates

- **Delete a template:** on screen 1, select it in the template list and press **Delete Template**.
- **Undo changes to a standard label, or bring back one you deleted:** on screen 1 press **Restore Standard Labels**. RECEIVING LABEL and the three ILSA labels return to their original layout and printer; your other templates stay as they are.
- **Bring back a single ILSA label** without touching the others: **Lookup Template** and choose its .rtm file (section 9).

> **Back up your templates:** Your adjusted templates live in `~/.local/share/ViewTouch/label_templates.json`. Copy that file somewhere safe after making changes you want to keep, or run the upload command in section 12.

11. Updating the program

New versions are published to your GitHub repository. To update the Pi:

- **Double-click the Update Label Utility icon** on the Pi's desktop (choose **Execute** if asked). A window opens, shows each step, beeps three times when it's finished, and says **Up to date**.
- Or, in a Terminal:

```
bash ~/viewtouch-main/modern/tools/update_label_utility.sh
```

Either way it: pulls the latest changes from GitHub (`main`), runs the installer, turns on the development-notes check, and updates GitHub's `main` to match the Pi. Then close and reopen the label utility. Your templates and printer settings are kept.

The installer sets up the launcher ViewTouch's Label Utility button uses, installs any missing Python parts, and installs Microsoft's Arial, Arial Black and Courier New fonts used by the ILSA labels. It may ask for the Pi's password. It can also be run on its own:

```
bash ~/viewtouch-main/modern/tools/install_label_utility.sh
```

> **If the update stops with an error:** nothing is lost. Take a photo of the window and send it to Bob.

12. Troubleshooting

Problem	What to do
Print says it was sent, but nothing prints	Check the Target line names the right printer. Then test the printer itself from a Terminal (replace the IP): `printf '^XA^FO40,40^A0N,50,50^FDTEST^FS^XZ' > /dev/tcp/192.168.1.24/9100` If TEST doesn't print, the printer is paused, out of labels/ribbon, open, or off.
Labels go to the wrong printer	Open Printer Offset or the designer and check Target. Check the printer's IP in ViewTouch's printer settings.
Top or bottom line cut off	Printer Offset: Print Higher or Print Lower, then Save & Test Print.
Right edge cut off	The template is wider than the labels. Reduce the label width in the designer (W × H, Apply Size).
Labels print too small or too large	Printer Resolution in Printer Offset doesn't match the printer (203 vs 300 dpi).
Labels skip, or print across two labels	New roll or different size: Printer Offset, Calibrate Gap.
Faint print, or bars run together	Raise or lower Darkness in Printer Offset.
Barcode won't scan	Check the right UPC was chosen (screen 3, UPC on Label). Make sure the barcode isn't cut off and Darkness is right.
Big price touches the barcode	The Arial Black font isn't installed. Run the installer (section 11).
An item is missing or has an old price	Press Reload viewtouch.db. Items marked inactive in ViewTouch, and restaurant menu items, are not listed.

Problem	What to do
Program won't open or closes by itself	Send the log (below) for diagnosis.

Sending the log for help

Right after a problem (before restarting the Pi), run:

```
bash ~/viewtouch-main/modern/tools/push_label_log.sh
```

This uploads the error log, the last label sent to a printer, and your current templates to the **pi-logs** branch of your GitHub repository, where they can be reviewed. It doesn't change any of your files.

13. Quick reference

To…	Do this
Open the label utility	ViewTouch Manager screen > Label Utility
Print shelf labels for some items	Retail Items > tick items > Next >> > Template > Print Labels > quantity > Print Now
Print labels for one brand	Tree View > double-click the brand > tick items > Next >> > Print Labels
Print receiving labels	Screen 1: double-click RECEIVING LABEL > Print Labels > quantity > Print Now
Print a carton or alternate UPC	Screen 3: Print UPC or Next UPC — or screen 4: UPC drop-down
Find an item by any of its UPCs	Screen 3: type or scan the code into Quick Search
Move the print up or down	Screen 4: Printer Offset > Print Higher / Lower > Save & Test Print
Set up a new label roll	Printer Offset > Calibrate Gap
Change a label's layout	Screen 4: Edit Report > drag pieces > Save
Make a new label from an existing one	Screen 4: Create Report > name, printer, size > Create & Open Designer
Bring in an ILSA label	Screen 1: Lookup Template > choose the .rtm file
See exact printer dots	Screen 4: double-click the preview
Delete a template	Screen 1: select it > Delete Template
Undo changes to the standard labels	Screen 1: Restore Standard Labels
Update the program	Double-click Update Label Utility on the desktop — or `bash ~/viewtouch-main/modern/tools/update_label_utility.sh`
Send logs for help	`bash ~/viewtouch-main/modern/tools/push_label_log.sh`